Chaitanya Uppalapati

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ABOUT

Data science graduate student at San Jose State University with a focus on AI/ML engineering. Experienced building agentic LLM applications and applied machine learning, including multi-agent systems, vision-based pipelines, and end-to-end models on real-world data. Comfortable across the stack, from model and agent design to data pipelines and cloud deployment. Seeking AI/ML and software engineering internships.

EDUCATION

San Jose State University — Jan 2025 – May 2027
M.S. in Applied Data Intelligence — *GPA 3.6 / 4.0*

KL University — 2020 – 2024
B.Tech in Computer Science — *GPA 3.84 / 4.0*

PROJECTS

Lighthouse: Multi-Agent AI Safety System — github.com/ChaitanyaUppalapati/LightHouse

- Built a multi-agent system on the Fetch.ai uAgents framework (Watcher, Guardian, and Escalation agents plus an ASI:One coordinator) that ingests signals, reasons with Claude Sonnet 4.6 at temperature 0, and routes every action through a deterministic gate that decides act-alone vs. escalate-to-human.
- Engineered the keystone safety gate as a rule-based classifier over a frozen action registry and confidence thresholds, guaranteeing reproducible act / ask / watch routing with fail-safe defaults on unknown or low-confidence inputs.
- Implemented a Browserbase and Stagehand computer-use executor that performs real, reversible actions on live web interfaces, plus genuine agent-to-agent orchestration over the Fetch.ai Chat Protocol with all agents registered on Agentverse.
- Instrumented the pipeline with Arize Phoenix tracing and an LLM-as-judge evaluator to measure and improve Watcher accuracy. Built for the UC Berkeley AI Hackathon 2026.

OpenPill: Medication-Safety Perception Pipeline — github.com/ChaitanyaUppalapati/OpenPill

- Designed the perception pipeline for a caregiver-facing medication-safety assistant on an “LLMs at the edges, determinism at the core” architecture, keeping the safety-critical decision out of any language model.
- Built a vision intake module using a Claude vision model to read pill-bottle images into structured medication hypotheses, gated by a deterministic composite-confidence ladder that caps partial reads below the auto-resolve threshold.
- Implemented a deterministic RxNorm normalization client that resolves hypotheses to canonical medications or routes low-confidence reads to a human confirmation request, enforcing a zero-silent-wrong-drug invariant verified by a pytest suite.

Hurricane Food Relief Priority: End-to-End ML — github.com/.../hurricane-food-relief-ml

- Built an end-to-end ML pipeline fusing 13 public data sources (FEMA, USDA, CDC SVI, Census ACS, NOAA IBTrACS) into a 5,500-row analytic base table to predict pre-landfall hurricane damage severity per ZIP code.
- Trained an XGBoost four-class severity classifier on a temporal train / validation / test split, reaching 0.78 macro ROC-AUC and 0.82 recall on the Severe class, with FEMA API pagination and EPSG:5070 equal-area spatial joins.
- Ran a Fairlearn equity audit confirming higher Severe-class recall for more vulnerable communities and shipped a SHAP-explained Streamlit dashboard ranking food-insecure ZIPs for relief logistics.

TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript, SQL, Java

AI / LLM Systems: Multi-agent orchestration (Fetch.ai uAgents, ASI:One), Anthropic Claude API (text and vision), LangGraph, RAG, LLM-as-judge evaluation, Browserbase / Stagehand, Pydantic

Machine Learning: scikit-learn, XGBoost, pandas, NumPy, SHAP, Fairlearn, HuggingFace, LoRA / PEFT

Data and Cloud: GCP, AWS, Apache Spark, Kafka, Docker, Kubernetes, Terraform, Postgres / pgvector, MongoDB, MySQL

Observability and Tools: Arize Phoenix, MLflow, Streamlit, pytest, Git, REST APIs